

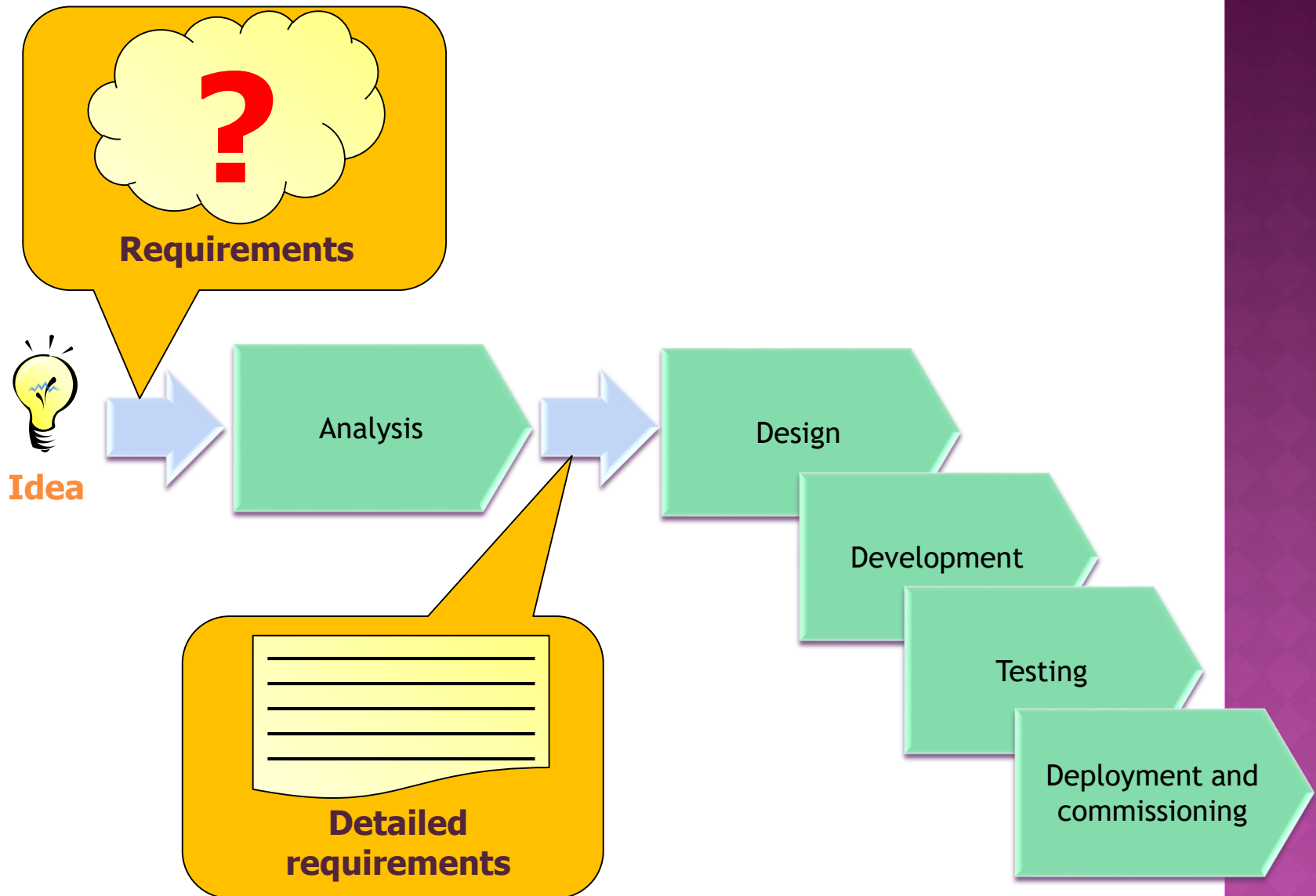
ANALYTICAL STAGE OF THE PROJECT

General Information

1 The place of the project's analytical stage in the lifecycle

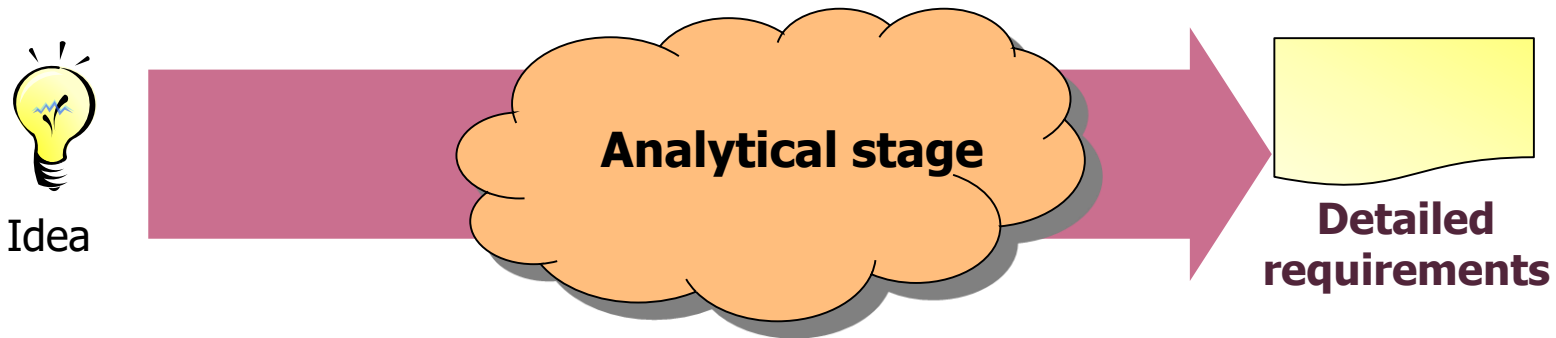
2 Structure of the project's analytical stage

THE PLACE OF THE ANALYTICAL STAGE IN THE LIFECYCLE



DEFINITION OF THE ANALYTICAL STAGE OF THE PROJECT

The analytical stage of the project consists of a sequence of activities aimed at identifying and documenting the requirements for the information system being designed.



The analysis phase of the project is the stage in the system development lifecycle where, through painstaking work and based on the client's disparate expectations and wishes, detailed requirements for the system formalized under design stage.

1 The place of the project's analytical stage in the lifecycle

2 Structure of the project's analytical stage

INPUT DATA FOR THE ANALYTICAL STAGE OF THE PROJECT

The input data for the analytical stage of the project consists of:

- **The Client's expectations and needs**
- **Specifics of the subject area and the Client's operations, including:**
 - departmental regulations
 - job descriptions and work instructions
 - methodological documents
 - legislation, state and industry standards, technical specifications
 - Other
- **Information regarding the Client's infrastructure**

KEY STAGES OF THE ANALYSIS

Identifying needs

- System goals and objectives
- Glossary of terms and definitions (agreed upon with the client)
- High-level processes (description of activities)
- IS context

Study of the activity to be automated (AS-IS)

- "As-Is" models (organizational, functional, informational, IT, etc.)
- Representation of current activities to be automated in the form of business processes

Identifying the needs of the Client and users

- Organized and structured needs of the Client and Users

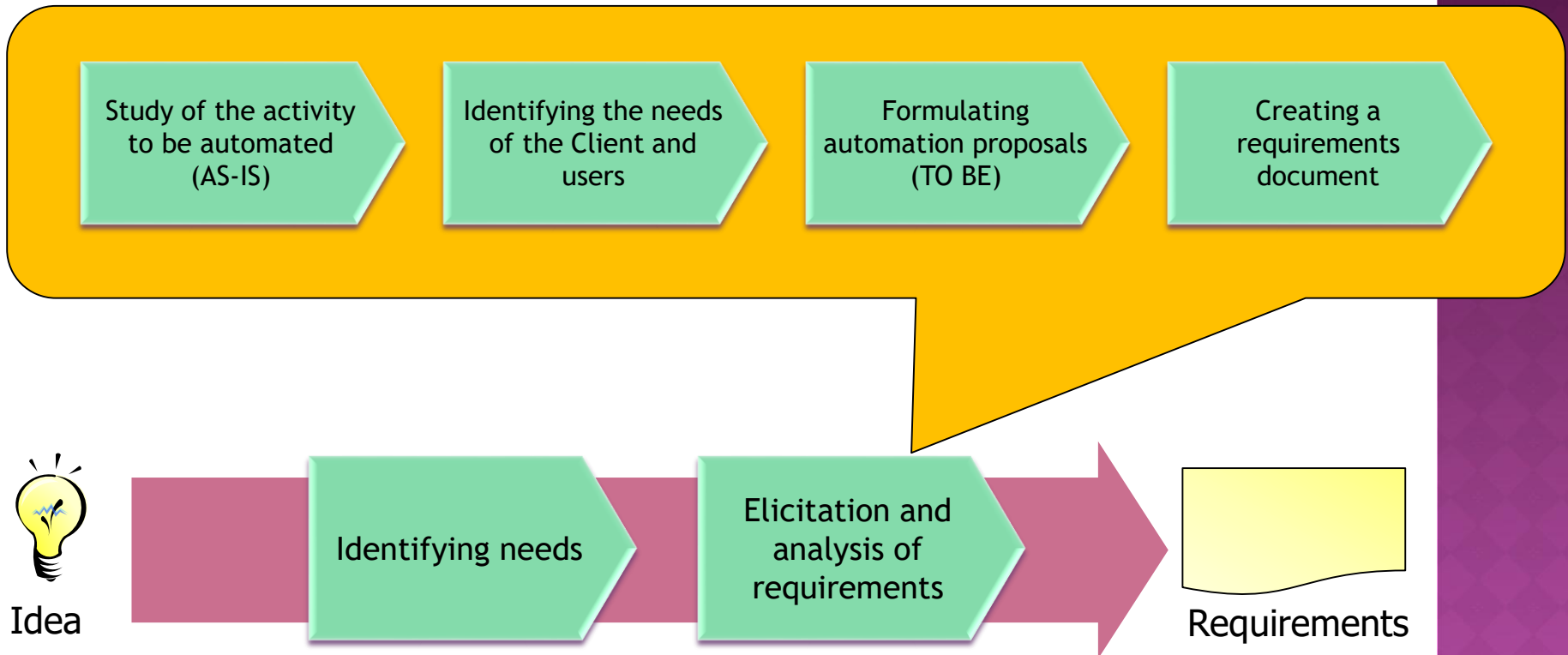
Formulating automation proposals(TO BE)

- Mapping requirements to current operations
- Representing the client's operations as a set of business processes modified to reflect the planned automation
- System use cases

Creating a requirements document

- Detailed requirements document

ANALYTICAL STAGE OF THE PROJECT



"IDENTIFYING NEEDS" STAGE

During the "Identifying Needs" stage, the analyst must:

- identify the goals the Client wishes to achieve through the system's implementation;
- define the scope of the system in sufficient detail to estimate the effort required for project implementation.

The results of the "Identifying Needs" stage serve as the basis for estimating the effort required to build the system.

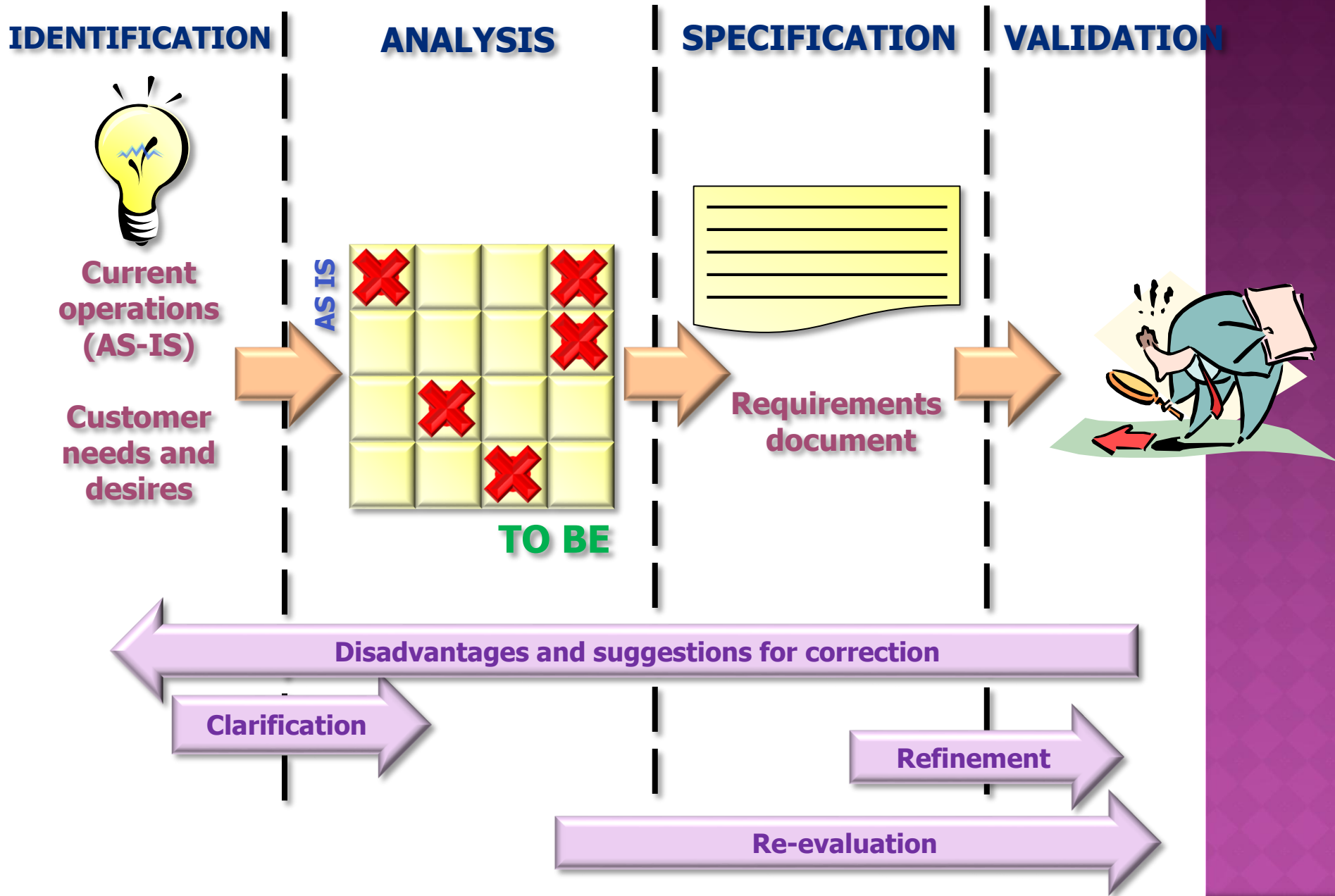
When implementing complex systems (or under conditions of high risk), it is virtually impossible to accurately estimate the total effort for the entire system development lifecycle. Therefore, in such cases, the system scope is used to estimate only the analysis stage, the results of which then serve as the basis for estimating the remaining project stages.

"REQUIREMENT IDENTIFICATION AND ANALYSIS" STAGE

During the "Requirement Identification and Analysis" stage, the analyst must:

- explore the Client's current operations (including infrastructure)—the "AS-IS" state
- identify, analyze, and structure the Client's expectations and needs
- map these needs against current operations and, based on the analysis, develop a vision of the "TO-BE" state (incorporating automation)
- formulate and document requirements for the System and its constituent software components

"REQUIREMENT IDENTIFICATION AND ANALYSIS" STAGE



"REQUIREMENT IDENTIFICATION AND ANALYSIS" STAGE

IDENTIFICATION

- Gathering information from stakeholders

ANALYSIS

- Processing and classifying the gathered information

SPECIFICATION

- Documenting information in the form of documents and diagrams

VALIDATION

- Confirming the documented information with users

RESULT OF THE PROJECT'S ANALYTICAL STAGE

The results of the project's analytical stage are:

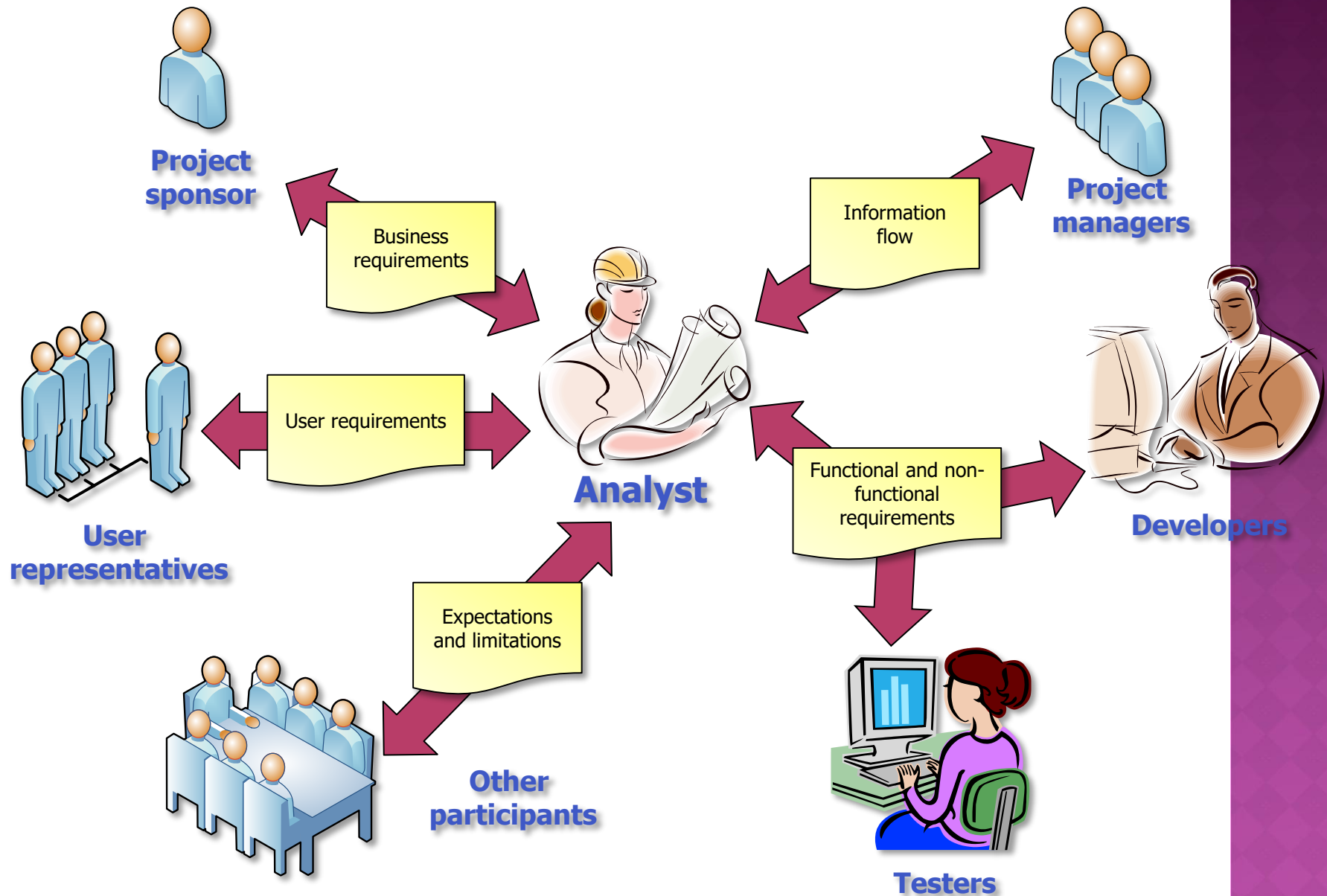
- **Business-level requirements:**
 - system goals and objectives
 - glossary of terms
 - information system application context
- **User-level requirements:**
 - "As-Is" model of the domain to be automated
 - ordered and structured user needs
 - "To-Be" model of the domain under automation
 - system use cases
- **Detailed requirements for the software components to be designed**

KEY CHARACTERISTICS OF THE RESULTS OF THE PROJECT'S ANALYTICAL STAGE

Requirements for the system and its software components must enable:

- **establishing a unified understanding of the system among all stakeholders**
- **carrying out system design and developing other design documentation**
- **monitoring the system's completeness status**
- **monitoring the system's alignment with the Customer's expectations**

ANALYST INTERACTION



ANALYST INTERACTION

Software developers and consumers share a common goal.

In information system development projects, tensions arise among analysts, developers, users, managers, and marketing specialists. The parties do not always trust the motivations of other participants, nor do they appreciate the needs and constraints of others.

The requirements analyst is the primary person responsible for ensuring close, cooperative, and positive collaboration with user representatives and other project stakeholders.

The analyst secures agreement on requirements from project participants, resulting in a win-win situation for everyone:

- clients are satisfied with the product;
- the development organization makes a profit;
- developers take pride in the work accomplished during a challenging project.

Describe the step-by-step actions you would take in each of the following situations:

- 1) Imagine you have been assigned to a project to automate the operations of JSC "NC Samruk-Kazyna." The project is at the contract-signing stage.
- 2) Imagine you have been assigned to a project to automate the operations of LLP "Aknur Travel." Three out of five components have already been implemented.